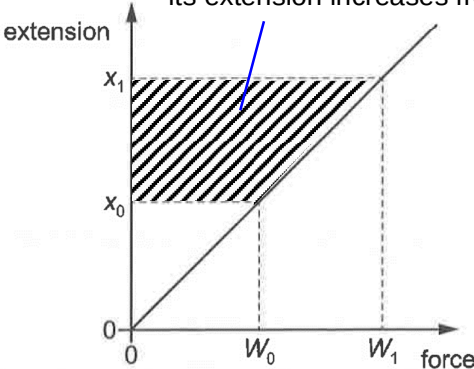


7	D	Elastic potential energy stored in a spring is equal to the area under the force-extension graph.

No.	Answer	Solution
		shaded area = extra elastic potential energy stored in the spring when its extension increases from x_0 to x_1  Elastic potential energy stored in spring when extension is x_0, $E_0 = \frac{1}{2} W_0 x_0$ Elastic potential energy stored in spring when extension is x_1, $E_1 = \frac{1}{2} W_1 x_1$ Extra elastic potential energy stored in spring when extension increases from x_0 to x_1, $\begin{aligned} \Delta E &= E_1 - E_0 \\ &= \frac{1}{2} W_1 x_1 - \frac{1}{2} W_0 x_0 \\ &= \frac{1}{2} (W_1 x_1 - W_0 x_0) \end{aligned}$
8	A	Time taken to travel 1000 m